

# Multiple Linear Regression

## STAT 223 – Applied Analytics

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# Simple Linear Regression

There are many cases where one variable is not enough to properly describe the behavior of a response variable.

For instance, if I want to predict the height of a newborn baby as an adult, I should probably consider *both* the mother's height and the father's height.

A **multiple linear regression** is a way to model the linear relationship between a response variable and  $p$  explanatory variables using a line drawn in  $(p + 1)$ -dimensional space known as a regression line.

# Multiple Linear Regression

The **population multiple linear regression** is defined as the equation:

$$\mathbb{E}[Y] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p,$$

where  $\mathbb{E}[Y]$  is the expected value of  $Y$ , given those values of the  $X_i$ 's.

# Multiple Linear Regression

The **multiple linear regression model** is defined as:

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon,$$

where  $\hat{Y}$  is the predicted value of  $Y$ , given the values of the  $X_i$ 's and  $\varepsilon \sim \text{NORMAL}(0, \sigma_e^2)$ , where  $\sigma_e^2$  is a constant variance across all values of  $X$ .

Note that, as  $\varepsilon \sim \text{NORMAL}(0, \sigma_e^2)$  and the vector  $\beta$ , the vector of the coefficients  $\beta_i$ , and  $\mathbf{X}$ , the data matrix, are “fixed”, then  $\hat{Y}$  is a random variable and  $\hat{Y} \sim \text{NORMAL}(\beta\mathbf{X}, \sigma_e^2)$ .

# Least Squares Regression

Obviously, we rarely, if ever, know the values for the entire population, so we can't ever truly know **beta** directly.

We have to estimate them from the sample values.

The **method of least squares** derives a linear regression model by minimizing the sum of squared errors:

$$\sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_{1i} - \beta_2 X_{2i} - \cdots - \beta_p X_{pi})^2$$

It can be shown that the estimates for  $\beta$ , denoted as **b**, can be found by:

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

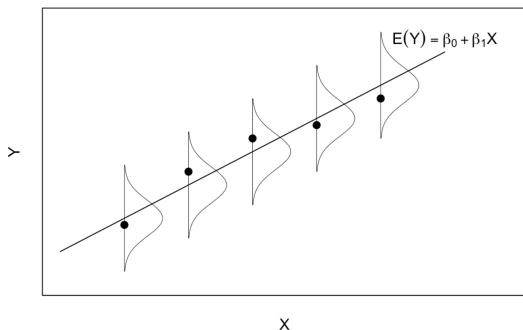
Recall, the difference between  $Y_i$ , the *actual* value, and  $\hat{Y}_i = b_0 + b_1 X_i$ , the predicted value of  $Y$  given  $X_i$ , is called the **residual**.

$$\varepsilon_i = Y_i - \hat{Y}_i$$

# Least Squares Regression

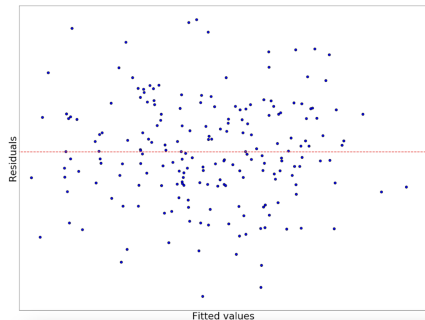
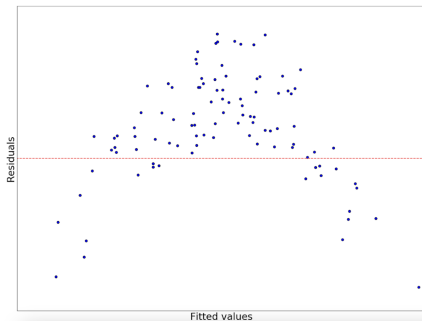
As with the simple linear regression, these residuals must satisfy:

1.  $\mathbb{E}[\varepsilon_i] = 0$ .
2.  $\sigma_e^2$  is the same for all  $x$ .
3.  $\varepsilon_i$  follows a NORMAL distribution.
4.  $\varepsilon_i \perp \varepsilon_j$  for  $i \neq j$ .



# Residual Plots

We can check these assumptions with residual plots, Q-Q plots and tests just like simple linear regression.



# Issues with Multiple Linear Regression

Some issues with multiple linear regression we need to consider:

- Outliers and High Leverage Observations
- The Curse of Dimensionality
- Multicollinearity

# Assessing Multiple Linear Regression

How do we determine if our model fits the data well? How can we determine if our model makes good predictions?

First we can look at the **coefficient of multiple determination**,  $R^2$ .

## Coefficient of Multiple Determination, $R^2$

The **coefficient of multiple determination**, denoted by  $R^2$ , measures the ratio of total variance in the response variable,  $Y$ , that is explained by the predictor variables  $X_1, X_2, \dots, X_p$  (explained by the model).

# Assessing Multiple Linear Regression

$$R^2 = \frac{SSR}{SST}$$

where:

$$SSR = \sum_{i=1}^N \left( \hat{Y}_i - \bar{Y} \right)^2$$

$$SSE = \sum_{i=1}^N \left( Y_i - \hat{Y} \right)^2$$

$$SST = SSR + SSE = \sum_{i=1}^N \left( Y_i - \bar{Y} \right)^2$$

# Residual Mean Square Error

## Residual Mean Square Error

**Residual Mean Squared Error** ( $RMSE$ ) is a metric used in regression analysis to assess how well a model fits the data, calculated by averaging the squared differences between predicted and actual values. A lower  $RMSE$  indicates a better fit, meaning the model's predictions are closer to the actual values.

$$RMSE = \sqrt{MSE} = \sqrt{\frac{SSE}{N - p}}$$

# ANOVA Table

You can get these from the ANOVA table:

| <b>Source</b> | <b><math>SS</math></b> | <b><math>df</math></b> | <b><math>MS</math></b> | <b><math>F</math></b> | <b><math>p</math></b> |
|---------------|------------------------|------------------------|------------------------|-----------------------|-----------------------|
| Model         | $SSR$                  | $df_R$                 | $MSR$                  | $F$                   | $P$                   |
| Error         | $SSE$                  | $df_E$                 | $MSE$                  |                       |                       |
| Total         | $SST$                  | $df_T$                 |                        |                       |                       |

# ANOVA Table

You can get these from the ANOVA table:

```
> m12 = ols('Y ~ Regression', fat).fit()  
> print(sm.stats.anova_lm(m12, typ=2))
```

|            | sum_sq     | df   | F         | PR(>F)   |
|------------|------------|------|-----------|----------|
| Regression | 389.455334 | 2.0  | 31.249317 | 0.000002 |
| Residual   | 105.934166 | 17.0 | NaN       | NaN      |

Each time you add new predictor variables into your model,  $R^2$  will either stay the same or increase, even if the additional input variables have no relationship with the output variables.  $R^2$  will never decrease, so it will always look like the expanded model is “better”.

To account for this, we use something called the **Adjusted  $R^2$** ,  $R_{adj}^2$ :

$$R_{adj}^2 = 1 - (1 - R^2) \left[ \frac{N - 1}{N - (k + 1)} \right].$$

$R_{adj}^2$  should only increase when a worthwhile predictor is added to the model.

# Interpretation of Coefficient Estimates

$b_0$  is what we would predict  $\hat{Y}$  to be when all the predictor variables  $X_i$  are zero.

If we keep all the predictor variables constant, and increase  $X_i$  by one unit, we would expect  $\hat{Y}$  to change by the value of  $b_i$ .  $\hat{Y}$  will increase if  $b_i$  is positive and decrease if  $b_i$  is negative.

## Multicollinearity

**Multicollinearity** is when two or more independent variables in a regression analysis are correlated, or when one predictor variable is a linear combination of other predictor variables.

The problem that multicollinearity bring is that if  $X_i$  and  $X_j$  are highly correlated, we increase  $X_i$  by one, even if we nominally keep  $X_j$  constant, there will still be implied changes due to the correlation.

# Checking for Multicollinearity

We can look at  $R^2$  for each predictor variable in terms of the other predictor variables. If we see a value of  $R^2 \geq 0.8$ , that may indicate a problem.

We can compare  $t$  and  $F$  statistics as well.

There are more advanced techniques like the *condition number* of the data matrix.

# Variance Inflation Factor

The **variance inflation factor**,  $VIF$ , is the reciprocal of  $1 - R^2$  for each predictor variable.

$$VIF = \frac{1}{1 - R^2}$$

If you see a  $VIF$  of greater than 5, you should start to worry. If you see a  $VIF$  of greater than 10, you should worry **a lot**.

# VIF and statsmodels

```
import pandas as pd
from statsmodels.stats.outliers_influence
    import variance_inflation_factor
from patsy import dmatrices

cars = pd.read_csv('https://data-analytics.zybooks.com/mtcars.csv')

y, X = dmatrices('mpg ~ cyl + disp + hp + drat + wt + qsec
                + vs + am + gear + carb',
                data=cars,
                return_type='dataframe')

# Calculate the VIF for each X_i
vif = [variance_inflation_factor(X.values, i+1)
        for i in range(X.shape[1]-1)]

print(vif)
```

# Confidence and Prediction Intervals

## Confidence Interval for the Mean

A **multiple linear regression confidence interval for the mean** is an interval around  $\hat{Y}$  that quantifies sampling uncertainty when  $\hat{Y}$  is used to estimate  $\mathbb{E}[Y]$  for fixed values of  $X_1, X_2, \dots, X_n$ .

## Prediction Interval for an Individual Response

A **multiple linear regression prediction interval for an individual response** is an interval around  $\hat{Y}$  that quantifies sampling uncertainty when  $\hat{Y}$  is used to predict  $Y$  for fixed values of  $X_1, X_2, \dots, X_n$ .

# Individual $t$ -Test

If we want to determine the significance of a single coefficient, we can do an individual  $t$ -test for just  $\beta_i$ .

$$H_0 : \beta_i = 0$$

$$H_A : \beta_i \neq 0$$

Remember that because

$Y \sim \text{NORMAL}(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p, \sigma_e^2)$ , “it can be shown” that the  $\beta_i$ ’s follow a NORMAL distribution themselves, so we can do a  $t$ -test.

$$T = \frac{b_i}{SE(b_i)} \sim t_{N-p-1}$$

Suppose we want to determine whether the model **as a whole** fits the data well and is significant, we use an  $F$  test.

$$H_0 : \beta_0 = \beta_1 = \beta_2 = \cdots = \beta_p = 0$$

$$H_A : \text{At least one of the } \beta_i \neq 0 \text{ for } i = 1, 2, \dots, p$$

The test statistic is:

$$F = \frac{MSR}{MSE} \sim F_{p-1, N-p}$$

Note that this is the same test statistic as in the ANOVA procedure.

We can also use the  $F$  test to test subsets of  $\beta_i$ 's, which we will discuss in a future lecture.